Avani Pillai

FPGA Design Engineer

RESUME_1769

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Total Experience: 4 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 4 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Python, ILA, VHDL, AXI4, SPI, FPGA, Tcl, Xilinx Vivado, Intel Quartus, Verilog, UART, DSP

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

FPGA Design Engineer | Foresight Industrial | Jan 2024 - Present

- Lowered end-to-end data-path latency by 32% using pipelined logic
- Improved maximum operating frequency by 22% after timing-path optimization
- Reduced FPGA resource utilization by 33% through arithmetic sharing
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate FPGA Design Engineer | JuniperWorks Systems | Jul 2022 - Dec 2023

- Reduced FPGA resource utilization by 34% through arithmetic sharing
- Verified 805 protocol corner cases with repeatable simulation suites

PROJECTS

High-Speed Data Logger | Xilinx Vivado, Python, ILA, VHDL

- Created DMA and buffering logic with ILA debugging hooks.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Real-Time Signal Filter | VHDL, FPGA, Xilinx Vivado, AXI4

- Implemented fixed-point FPGA pipelines with testbench golden models.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Sahyadri College of Engineering, 2022
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

FPGA Architecture Workshop (2022)